
Nandhini Anand

Senior AI Architect & Data Strategy Lead

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Bangalore, India

PROFESSIONAL SUMMARY

Senior AI Architect & Data Engineering Lead with 15+ years of experience bridging the gap between Data Science and robust Software Engineering. Expert in designing end-to-end Machine Learning systems—from raw sensor data ingestion (IoT/Embedded) to scalable cloud deployment and visualization. Proven track record of modernizing legacy pipelines (reducing latency from hours to minutes), leading technical teams, and deploying production-grade AI solutions using Python, Kafka, and Deep Learning. Passionate about building systems that are not just theoretically sound, but operationally resilient.

TECHNICAL SKILLS

- **Core Languages:** Python (Expert), Rust, C, Assembly, SQL, Bash
- **IoT & Edge Computing:** Raspberry Pi, Edge AI Deployment, Sensor Data Integration, MQTT/IoT Protocols, Hardware-Software Interfacing
- **AI & Machine Learning:** Computer Vision (CNNs, Object Detection), Generative Models (GANs), NLP, Deep Learning (TensorFlow/PyTorch), Scikit-learn
- **Data Engineering:** Kafka, Storm, Airflow, Hadoop Ecosystem, Redis, PostgreSQL, MongoDB, Cassandra
- **Cloud & DevOps:** AWS, Azure, GCP, DigitalOcean, Docker, CI/CD
- **Visualization:** D3.js, Bokeh, Seaborn, Plotnine

PROFESSIONAL EXPERIENCE

Independent AI Consultant & Solutions Architect | Remote

Oct 2023 – Present

- **Smart City Computer Vision System:** Architected and deployed an end-to-end edge AI solution for civic monitoring (illegal dumping detection). Built a custom pipeline using Raspberry Pi + GPU acceleration to process video feeds in real-time. Implemented object detection (CNNs) and automated reporting mechanisms (Number Plate Recognition).
- **Green Energy Optimization Engine:** Refactored and optimized a legacy decision engine for solar panel layout recommendations. Improved computational efficiency for a local policy platform, enabling faster user feedback loops.

- **Technical Education:** Designed comprehensive curricula on Data Storytelling and Python/SQL for analytics. Conducted intensive workshops for junior engineers, bridging the gap between academic theory and production coding standards.

The Math Company | *Senior Data Scientist (Contract)*

Dec 2022 – Oct 2023

- **Pipeline Modernization:** Led the transformation of the NBA (Next Best Action) pipeline from batch Python scripts to near-real-time SQL execution. **Reduced latency from hours to minutes** and improved conversion rates by **50%** (1% to 1.5%).
- **Workforce Analytics:** Delivered a searchable dashboard to track skill distribution and project allocation for 800+ employees.
- **Training:** Conducted faculty training at SRM and SASTRA Universities as part of a data science enablement initiative.

VIT Infotech | *Senior Technical Architect (Contract)*

Mar 2022 – Oct 2022

- **Stream Processing:** Implemented Faust and Kafka stream processing for real-time data handling.
- **Sensor Fusion:** Integrated on-site code modules to process GPS and pressure sensor data for precise altitude estimation.
- **Decision Systems:** Deployed a complex decision-maker app to validate calibration attempts from mobile devices.
- **Mentorship:** Guided junior engineers on best coding practices, debugging, and environment setup.

Mouritech Pvt. Ltd | *Senior Technical Architect (Contract)*

Mar 2021 – Sep 2021

- **Search Infrastructure:** Contributed to a search engine parsing multiple document formats (PDF, DOCX) for indexing and retrieval.
- **Azure Integration:** Built exception and validation modules for an SAP invoice processing application using Azure Functions.
- **Team Leadership:** Led a team of 5 engineers, establishing standards for development, demo, and production environments using Docker.

Smartek21 Pvt. Ltd. | *Senior Technical Specialist*

Oct 2020 – Feb 2021

- **Intelligent Document Processing (IDP):** Architected and led the development of a generic Q&A Chatbot engine capable of ingesting arbitrary PDF documents.
- **NLP & Parsing:** Utilized classical ML techniques and custom parsers to extract structured content from unstructured text, enabling natural language querying of document knowledge bases.
- **Team Leadership:** Assembled and directed the engineering team to deliver the MVP.

Trishula Adv. Comp. and Electronics | *Senior Data Scientist*

Apr 2019 – Jul 2020

- **IoT Data Pipelines:** Assembled a lightweight framework for collecting sensor/device status data from distributed IoT devices.
- **Simulation Modeling:** Built a dataset simulator and model to analyze the impact of carbon tax incentives on green energy market demand.
- **Code Quality:** Established code conventions and led peer review processes to improve team output.

Kaliber Labs Pvt. Ltd | *Senior Data Scientist*

Apr 2017 – Mar 2019

- **Big Data Analytics:** Assembled a text analytics pipeline leveraging STORM, Kafka, and Hadoop for parsing massive XML datasets.
- **Audio/Video AI:** Built NN-based models for audio cleaning (GANs) and video analytics for emotion prediction.
- **Speech-to-Text:** Implemented a Kaldi-based pipeline after evaluating Azure, Google Cloud, and IBM Watson alternatives.
- **Medical Tech:** Built a kidney stone size estimation tool for lithotripsy procedures using Paperspace cloud.

Greytip HR Solutions | *Senior Data Scientist*

Aug 2015 – Mar 2017

- **SaaS Analytics:** Deployed dashboards for monitoring SaaS metrics (MRR, LTV, CAC) and attrition health.
- **Predictive Modeling:** Built Random Forest models to predict customer attrition, directly enabling proactive engagement to reduce churn.
- **Data Warehouse:** Designed and set up the end-to-end data warehouse, ETL pipelines, and data sanity checks.

Early Career & Startup Experience | *Various Technical Roles*

2009 – 2015

- **Scale & Performance:** Engineered backend systems to handle growth from **1 Million to 2 Million users**, migrating from MongoDB/AWS to Redis/GCS.
- **Automation:** Built a video annotation tool that **reduced manual effort by 50%**.
- **User Retention:** Designed Cohort Analysis tools and engagement algorithms (TF-IDF ranking) to drive product features.
- **Chatbots:** Built early NLP-based gaming chatbots to increase user time-in-app.

EDUCATION

- **M.Sc.** | Allahabad University (2009)
- **B.E.** | Madurai Kamaraj University (2004)